A. Attention Matching Details

This appendix restates the attention-matching objectives from Section 2 in more detail.

A.1. Notation

Fix a single layer and KV-head with head dimension d. Let the original keys and values be

$$\boldsymbol{K} \in \mathbb{R}^{T \times d}, \qquad \boldsymbol{V} \in \mathbb{R}^{T \times d},$$

and let the compacted keys, values, and per-token bias be

$$\boldsymbol{C}_k \in \mathbb{R}^{t \times d}, \qquad \boldsymbol{C}_v \in \mathbb{R}^{t \times d}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{t}.$$

Let $\boldsymbol{Q}_{\text{ref}} \in \mathbb{R}^{n \times d}$ denote the reference query matrix whose ith row is $\boldsymbol{q}_i$.

Attention operators. For any key/value block $(\boldsymbol{K}, \boldsymbol{V})$ and query $\boldsymbol{q}$, define the scaled logits

$$\ell(\boldsymbol{q}; \boldsymbol{K}) = \tfrac{1}{\sqrt{d}} \boldsymbol{q} \boldsymbol{K}^\top$$

and the unnormalized mass

$$\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}) \;=\; \sum_j \exp(\ell(\boldsymbol{q}; \boldsymbol{K})_j).$$

Define the locally normalized attention output

$$\mathrm{Attn}(\boldsymbol{q}; \boldsymbol{K}, \boldsymbol{V}) \;=\; \frac{\exp(\ell(\boldsymbol{q}; \boldsymbol{K}))\, \boldsymbol{V}}{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K})}.$$

For compacted parameters $(\boldsymbol{C}_k, \boldsymbol{\beta}, \boldsymbol{C}_v)$ we use

$$\ell(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}) = \tfrac{1}{\sqrt{d}} \boldsymbol{q} \boldsymbol{C}_k^\top + \boldsymbol{\beta}, \quad \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}) = \sum_j \exp(\ell(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta})_j),$$

and

$$\mathrm{Attn}(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}, \boldsymbol{C}_v) = \frac{\exp(\ell(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}))\, \boldsymbol{C}_v}{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta})}.$$

A.2. Mass-preserving Attention Matching

We justify replacing attention over $[\boldsymbol{K}; \boldsymbol{K}_{\text{fixed}}]$ with the two objectives (1)–(2). However, we observe that attention over 2 subsets of indices decomposes into a mixture whose weights are determined by *unnormalized attention mass*.

Let $(\boldsymbol{K}, \boldsymbol{V})$ be the block we compact and $(\boldsymbol{K}_{\text{fixed}}, \boldsymbol{V}_{\text{fixed}})$ be any other block (e.g., chat-template tokens and/or future tokens). For any query $\boldsymbol{q}$,

$$\begin{aligned}\mathrm{Attn}\left(\boldsymbol{q}; \begin{bmatrix} \boldsymbol{K} \\ \boldsymbol{K}_{\text{fixed}} \end{bmatrix}, \begin{bmatrix} \boldsymbol{V} \\ \boldsymbol{V}_{\text{fixed}} \end{bmatrix}\right) = {} & \frac{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K})}{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}) + \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}_{\text{fixed}})} \mathrm{Attn}(\boldsymbol{q}; \boldsymbol{K}, \boldsymbol{V}) \\ & + \frac{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}_{\text{fixed}})}{\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}) + \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}_{\text{fixed}})} \mathrm{Attn}(\boldsymbol{q}; \boldsymbol{K}_{\text{fixed}}, \boldsymbol{V}_{\text{fixed}}).\end{aligned} \tag{5}$$

Now suppose we replace $(\boldsymbol{K}, \boldsymbol{V})$ by compact parameters $(\boldsymbol{C}_k, \boldsymbol{\beta}, \boldsymbol{C}_v)$. If for queries of interest we ensure

$$\mathrm{Attn}(\boldsymbol{q}; \boldsymbol{K}, \boldsymbol{V}) \approx \mathrm{Attn}(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}, \boldsymbol{C}_v) \quad \text{and} \quad \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K}) \approx \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{C}_k, \boldsymbol{\beta}),$$

then the mixture identity (5) implies that the full attention output is preserved even when *arbitrary* $\boldsymbol{K}_{\text{fixed}}, \boldsymbol{V}_{\text{fixed}}$ are appended later, because both mixture weights and the compacted-block contribution match. This means we can use objectives (1)–(2) to do one-shot compaction without knowing future keys/values.

Why biases matter. If $\boldsymbol{C}_k$ is chosen as a subset of $\boldsymbol{K}$ and we do not use attention biases, then $\mathrm{Mass}(\boldsymbol{q}; \boldsymbol{C}_k) \leq \mathrm{Mass}(\boldsymbol{q}; \boldsymbol{K})$ for every $\boldsymbol{q}$, so the compacted block systematically receives too little global weight in (5). A bias $\boldsymbol{\beta}$ introduces multiplicative weights $\exp(\boldsymbol{\beta}_j)$ so that each retained key can represent the mass of many removed keys, making mass matching feasible. Furthermore, even if $\boldsymbol{C}_k$ is not chosen as a subset of $\boldsymbol{K}$, there are cases where the bias is necessary to match attention mass, such as $\boldsymbol{q} = \boldsymbol{0}$, which requires matching T vs t. For these reasons, introducing attention biases after compaction is very natural.

Stable computation. In implementation, we evaluate Mass and Attn using the standard per-query max-shift for numerical stability. For logits ℓ with $s = \max_j \ell_j$, we compute $\sum_j \exp(\ell_j) = \exp(s) \sum_j \exp(\ell_j - s)$ and similarly for the numerator of Attn. This does not change any of the identities above. For OMP key selection, we operate on shifted features $\exp(\ell - s)$ and shifted targets $\sum_j \exp(\ell - s)$.

B. Experimental Details

B.1. Models

We evaluate `Qwen3-4B` (Qwen Team, 2025), `Llama-3.1-8B-Instruct` (Llama Team, 2024), and `Gemma-3-12b-it` (Gemma Team, 2025). We use a maximum generation length of 2048 tokens and each model's default decoding settings (temperature, top-k, and top-p). For long-context benchmarks (LongHealth), we use the `Qwen3-4B-Instruct-2507` variant instead of `Qwen3-4B` because it has a longer native sequence length.

B.2. Datasets

Our default evaluation dataset is **QuALITY** (Pang et al., 2022), a question-answering dataset testing comprehension of 5-7k-token passages. For our evaluations, we first compact the article context and then evaluate performance on the resulting QA task by batched decoding over the associated questions. We believe that pairing each article with many comprehension-focused multiple-choice questions makes QuALITY a controlled and informative benchmark for compaction.

We evaluate performance over the first 50 articles in the validation set (894 questions). Due to the computational cost of Cartridges ($\sim$5 hours per context), we only evaluate it on Qwen3-4B, on the first 20 articles (360 questions). Original-cache accuracy on this subset (75.0%) slightly exceeds that of the full set (72.1%).

We evaluate chunked compaction on **LongHealth** (Adams et al., 2024) by concatenating 5 patient records per context. This yields 4 contexts of roughly 60k tokens each, with 100 multiple-choice questions per context. We compact each context in 5 chunks before evaluating QA performance. While Eyuboglu et al. (2025) concatenate 10 patient records, we found that our evaluated base models' performance degraded beyond $\sim$100k tokens, which we attribute to limitations in long-context extrapolation of the model rather than the compaction procedure. In contrast, Cartridges optimizes a single latent representation across chunks and is thus not sensitive to long-context brittleness in the base model.

We show an example context snippet and question below for QuALITY and LongHealth. These tasks can involve heavy information-extraction and reasoning across the long context, making them an ideal and difficult testbed for compaction.

QuALITY

Context (excerpt):
THE GIRL IN HIS MIND By ROBERT F. YOUNG
Transcriber's Note: This etext was produced from Worlds of Tomorrow April 1963
Extensive research did not uncover any evidence that the U.S. copyright on this publication was renewed.
Every man's mind is a universe with countless places in which he can hide—even from himself!
... (5-7k token article)

Sample Question:
Why does Deirdre get so upset when Blake Past suggests she go to prom with the young man?
A) Because Blake is trying to guilt Deirdre into going with the young man by telling her that it'll ease her conscience.
B) Because Deirdre has fallen in love with Blake, despite his age, and wants him to take her to the prom.
C) Because Blake is acting like he's her father, which is a sensitive topic for Deirdre because she lost her real parents.
D) Because the young man gave up his right arm in order to afford tickets to the prom, and this disgusts Deirdre.

LongHealth

Context (excerpt):
Dear colleague,
We wish to provide an update regarding Mrs. Anna Sample, born on 01.01.1970. She was admitted to our clinic from 01/01/2017 to 01/02/2017.
Diagnosis: Diffuse large B-cell lymphoma of germinal center type; ID 01/2017
- Ann-Arbor: Stage IV
- R-IPI: 2 (LDH, stage)
- CNS-IPI: 2
- Histology: Aggressive B-NHL (DLBCL, NOS); no evidence of t(14;18) translocation. Ki-67 at 40%. Positive reaction to MUM1, numerous CD68-positive macrophages. Negative reaction to ALK1 and TdT.
- cMRI: Chronic inflammatory lesions suggestive of Multiple Sclerosis (MS)
- CSF: no evidence of malignancy
- Bone marrow aspiration: no infiltration from the pre-existing lymphoma.
Current treatment:
Initiated R-Pola-CHP regimen q21
- Polatuzumab vedotin: 1.8mg/kg on Day 1.
... (10-12k tokens per patient; 5 patients)

Sample Question:
Mrs. Sample received multiple radiologic examinations. In which order did she receive them?
A) MR spine > MR Head > CT Thoracic Spine > CT Whole Body
B) MR Brain > MR thoracic/lumbar spine > CT chest/abdomen/pelvis > CT Thoracic Spine > CT Thoracic Spine > CT chest/abdomen/pelvis > Abdominal ultrasound > MR Spine > Whole-body PET/CT > Liver MRI
C) MR cervical spine > MR Head > CT Whole Body > CT Thoracic Spine
D) MR Brain > CT Thoracic Spine > MR cervical spine > CT Whole Body
E) CT Thoracic Spine > CT Whole Body > MR cervical spine > MR Head

B.3. Baselines

Cartridges. We run experiments using the Cartridges repository (Eyuboglu et al., 2025). For each QuALITY article, we synthesize 32,768 self-study samples (25–30M tokens) using the default settings and train for 1 epoch with the default learning rate (2×10^{-2}). We replicate the same order of magnitude of training compute for a fair comparison (Table 3). We use the same evaluation setup as in our other experiments. Optimizing training or using more diverse self-study samples may greatly speed up results; we use the default settings provided in the open-source repository.

Dataset	Context	Self-Study Tokens	H100-hours
LongHealth-5	60k	100–150M	$\sim$15
Quality	7k	25–30M	$\sim$5
LongHealth-10 (Original)	120k	300M	-

Table 3. **Training compute for Cartridges.** We sample approximately the same number of self-study tokens per context token as in the original Cartridges implementation.

H2O, SnapKV, PyramidKV, and KVzip. We evaluate H2O, SnapKV, PyramidKV, and KVzip (Zhang et al., 2023; Li et al., 2024; Cai et al., 2025; Kim et al., 2025b). These methods select a subset of keys based on highest attention scores under a chosen set of queries.

H2O was originally proposed as an online eviction policy; here we use its one-shot compaction analogue, which we term **H2O+**. Concretely, during context prefill we collect all queries for each KV-head position and then select the top-k tokens per head by attention score, rather than evicting tokens sequentially based on only partial query sets. Moreover, because we compact only the content portion of the context—keeping chat-template tokens and the question portion fixed—we do not explicitly retain all recent tokens as in the original paper. Finally, we use the GQA variant of this method (and of the